United Nations

Our Common Future
Chapter 6: Species and Ecosystems: Resources for Development

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I. The Problem: Character and Extent

II. Extinction Patterns and Trends

III. Some Causes of Extinction

IV. Economic Values at Stake

V. A New Approach; Anticipate and Prevent

VI. International Action for National Species

1. Some Current Initiatives
2. Setting Priorities

VII. Scope for National Action

VIII. The Need for Action

Footnotes

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1. Conservation of living natural resources - plants, animals, and micro-organisms, and the non-living elements of the environment on which they depend - is crucial for development. Today, the conservation of wild living resources is on the agenda of governments; nearly 4 per cent of the Earth's land area is managed explicitly to conserve species and ecosystems, and all but a small handful of countries have national parks. The challenge facing nations today is no longer deciding whether conservation is a good idea, but rather how it can be implemented in the national interest and within the means available in each country.

I. The Problem: Character and Extent

2. Species and their genetic materials promise to play an expanding role in development, and a powerful economic rationale is emerging to bolster the ethical, aesthetic, and scientific cases for preserving them. The genetic variability and germplasm material of species make contributions to agriculture, medicine, and industry worth many billions of dollars per year.

3. Yet scientists have intensively investigated only one in every 100 of Earth's plant species, and a far smaller proportion of animal species. If nations can ensure the survival of species, the world can look forward to new and improved foods, new drugs and medicines, and new raw materials for industry. This - the scope for species to make a fast-growing contribution to human welfare in myriad forms - is a major justification for expanded efforts to safeguard Earth's millions of species.

4. Equally important are the vital life processes carried out by nature, including stabilization of climate, protection of watersheds and soil, preservation of nurseries and breeding grounds, and so on. Conserving these processes cannot be divorced from conserving the individual species within natural ecosystems. Managing species and ecosystems together is clearly the most rational way to approach the problem. Numerous examples of workable solutions to local problems are available./1

5. Species and natural ecosystems make many important contributions to human welfare. Yet these very important resources are seldom being used in ways that will be able to meet the growing pressures of future high demands for both goods and services that depend upon these natural resources.

6. There is a growing scientific consensus that species are disappearing at rates never before witnessed on the planet. But there is also controversy over those rates and the risks they entail. The world is losing precisely those species about which it knows nothing or little; they are being lost in the remotest habitats. The growing scientific concern is relatively new and the data base to support it fragile. But it firms yearly with each new field report and satellite study.

7. Many ecosystems that are rich biologically and promising in material benefits are severely threatened. Vast stocks of biological diversity are in danger of disappearing just as science is learning how to exploit genetic variability through the advances of genetic engineering. Numerous studies document this crisis with examples from tropical forests, temperate forests, mangrove forests, coral reefs, savannas, grasslands, and arid zones./2 Although most of these studies are generalized in their documentation and few offer lists of individual species at risk or recently extinct, some present species-by-species details. (See Box 6-1.)

8. Habitat alteration and species extinction are not the only threat. The planet is also being impoverished by the loss of races and varieties within species. The variety of genetic riches inherent in one single species can be seen in the variability manifested in the many races of dogs, or the many specialized types of maize developed by breeders./3

9. Many species are losing whole populations at a rate that quickly reduces their genetic variability and thus their ability to adapt to climatic change and other forms of environmental adversity. For example, the remaining gene pools of major crop plants such as maize and rice amount to only a fraction of the genetic diversity they harboured only a few decades ago, even though the species themselves are anything but threatened. Thus there can be an important difference between loss of species and loss of gene reservoirs.

10. Some genetic variability inevitably will be lost, but all species should be safeguarded to the extent that it is technically, economically, and politically feasible. The genetic landscape is constantly changing through evolutionary processes, and there is more variability than can be expected to be protected by explicit government programmes. So in terms of genetic conservation, governments must be selective, and ask which

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gene reservoirs most merit a public involvement in protective measures. However, as a more general proposition, governments should enact national laws and public policies that encourage individual, community, or corporate responsibility for the protection of gene reservoirs.

11. But before science can focus on new ways to conserve species, policy makers and the general public for whom policy is made must grasp the size and the urgency of the threat. Species that are important to human welfare are not just wild plants that are relatives of agricultural crops, or animals that are harvested. Species such as earthworms, bees, and termites may be far more important in terms of the role they play in a healthy and productive ecosystem. It would be grim irony indeed if just as new genetic engineering techniques begin to let us peer into life's diversity and use genes more effectively to better the human condition, we looked and found this treasure sadly depleted.

Box 6-1
Some Examples of Species Extinction

- In Madagascar, until about mid-century, there were 12,000 plant species and probably around 190,000 animal species, with at least 60 per cent of them endemic to the island's eastern strip of forest (that is, found nowhere else on Earth). At least 93 per cent of the original primary forest has been eliminated. Using these figures, scientists estimate that at least half the original species have already disappeared, or are on the point of doing so.

- Lake Malawi in Central Africa holds over 500 cichlid fish species, 99 per cent of them endemic. The lake is only one-eighth the size of North America's Great Lakes, which feature just 173 species, fewer than 10 per cent of which are endemic. Yet Lake Malawi is threatened through pollution from industrial installations and the proposed introduction of alien species.

- Western Ecuador is reputed to have once contained between 8,000 and 10,000 plant species, some 40 and 60 per cent of them endemic. Given that there are between 10 and 30 animal species for every one plant species in similar areas, western Ecuador must have contained about 200,000 species. Since 1960, almost all the forests of western Ecuador have been destroyed to make way for banana plantations, oil wells, and human settlements. The number of species thus eliminated is difficult to judge, but the total could well number 50.000 or more - all in just 25 years.

- The Pantanal area of Brazil contains 110,000 square kilometres of wetlands, probably the most extensive and richest in the world. They support the largest and most diversified populations of waterfowl in South America. The area has been classified by UNESCO as 'of international importance'. Yet it suffers increasingly from agricultural expansion, dam construction, and other forms of disruptive development. Sources: W. Rauh. 'Problems of Biological Conservation in Madagascar', in D. Bramwell (ed.), Plants and Islands (London: Academic Press, 1979): D.C.N. Barel et al., 'Destruction of Fisheries in Africa's Lakes', Nature, Vol. 315, pp. 19-20, 1985; A.H. Gentry. 'Patterns of Neotropical Plant Species Diversity', Evolutionary Biology. Vol. 15, pp.1-84, 1982; D.A. Scott and M. Carbonell, 'A Directory of Neotropical Wetlands', IUCN. Gland. Switzerland. 1985.

Our Atlantic forest, this mass of tropical forest that is a narrow stretch from the North to the South, has been reduced drastically.

This forest is characterised by a large number of endemic species, that is species that only exist in this area, and only exist in Brazil. And consequently, it is up to us, Brazilians, to shoulder the responsibility of keeping these species in existence.

Ibsen de Gusmao Camara
President, Brazilian Foundation for Preservation of Nature
WCED Public Hearing
Sao Paulo. 28-29 Oct 1985

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II. Extinction Patterns and Trends

12. Extinction has been a fact of life since life first emerged. The present few million species are the modern-day survivors of the estimated half-billion species that have ever existed. Almost all past extinctions have occurred by natural processes, but today human activities are overwhelmingly the main cause of extinctions.

13. The average duration of a species is some 5 million years. The best current estimates are that on average 900,000 species have become extinct every 1 million years during the last 200 million years, so the average 'background rate' of extinction has been very roughly one in every one and one-ninth years./4 The present human-caused rate is hundreds of times higher, and could easily be thousands of times higher./5 We do not know. We have no accurate figures on the current rates of extinctions, as most of the species vanishing are those least documented, such as insects in tropical forests.

14. Although tropical moist forests are by far the richest biological units in terms of genetic diversity and by far the most threatened through human activities, other major ecological zones are also under pressure. Arid and semi-arid lands harbour only a very small number of species compared with tropical forests. But because of the adaptations of these species to harsh living conditions, they feature many potentially valuable biochemicals, such as the liquid wax of the jojoba shrub and the natural rubber of the guayule bush. Many of these are threatened by. among other things, the expansion of livestock herding.

15. Coral reefs, with an estimated half-million species in their 400,000 square kilometres, are being depleted at rates that may leave little but degraded remnants by early next century. This would be a great loss, in that coral-reef organisms, by virtue of the 'biological warfare' they engage in to ensure living space in crowded habitats, have generated an unusual number and variety of toxins valuable in modern medicine./6

16. Tropical moist forests cover only 6 per cent of the Earth's land surface but contain at least half the Earth's species (which totals 5 million at a minimum, but could be as many as 30 million). They may contain 90 per cent or even more of all species. The mature tropical forests that still exist cover only 900 million hectares, out of the 1.5-1.6 billion hectares that once stood. Between 7.6 million and 10 million hectares are eliminated outright each year, and at least a further 10 million hectares are grossly disrupted annually./7 But these figures come from surveys of the late 1970s, and since then deforestation rates have probably accelerated.

17. By the end of the century, or shortly thereafter, there could be little virgin tropical moist forest left outside of the Zaire Basin and the western half of Brazilian Amazonia, plus some areas such as the Guyana tract of forest in northern South America and parts of the island of New Guinea. The forests in these zones are unlikely to survive beyond a few further decades, as world demand for their produce continues to expand and as the number of forestland farmers increases.

18. If deforestation were to continue in Amazonia at present rates until the year 2000, but then halted completely (which is unlikely), about 15 per cent of plant species would be lost. Were Amazonia's forest cover to be ultimately reduced to those areas now established as parks and reserves, 66 per cent of plant species would eventually disappear, together with almost 69 per cent of bird species and similar proportions of all other major categories of species. Almost 20 per cent of the Earth's species are found in Latin American forests outside of Amazonia; another 20 per cent are found in forests of Asia and Africa outside the Zaire Basin./8 All these forests are threatened, and if they were to disappear, the species loss could amount to hundreds of thousands.

19. Unless appropriate management measures are taken over the longer term, at least one-quarter, possibly one-third, and conceivably a still larger share of species existing today could be lost. Many experts suggest that at least 20 per cent of tropical forests should be protected, but to date well under 5 per cent has been afforded protection of any sort - and many of the tropical forest parks exist only on paper.

20. Even the most effectively managed parks and protected areas are unlikely to provide a sufficient answer. In Amazonia, if as much as half the forest were to be safeguarded in some way or another but the other half were to be eliminated or severely disrupted, there might well not be enough moisture in the Amazonian ecosystem to keep the remaining forest moist./9 It could steadily dry up until it became more like an open woodland - with the loss of most of the species adapted to tropical moist forest conditions.

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21. More widespread climatic changes are likely to emerge within the foreseeable future as the accumulation of 'greenhouse gases' in the atmosphere leads to global warming early in the next century. (See Chapter 7.) Such a change will produce considerable stress for all ecosystems, making it particularly important that natural diversity be maintained as a means of adaptation.

Twenty years ago, as we decided to intensify our forest exploitation, we just thought the resource is available, and we just took it. At the time, we also thought the intensive selecting out of the trees being cut wouldn't destroy forest regeneration. Because not all of the trees were being cut. But we forgot that we don't know yet about how the tropical forest should be rehabilitated.

An indigenous species such as meranti, I don't know the name in English, meranti, rami, is our high-valued wood, a timber that cannot make a shadow in its particular period of growth. And it cannot survive without that shadow. And we still didn't think about it, we just accepted the technology from the West that we have to cut, to exploit our forest.

Emmy H. Dharsono
NGO Network for Forest Conservation
WCED Public Hearing
Jakarta, 26 March 1985

III. Some Causes of Extinction

22. The tropics, which host the greatest number and diversity of species, also host most developing nations, where population growth is fastest and poverty is most widespread. If farmers in these countries are forced to continue with extensive agriculture, which is inherently unstable and leads to constant movement, then farming will tend to spread throughout remaining wildlife environments. But if they are helped and encouraged to practise more intensive agriculture, they could make productive use of relatively limited areas, with less impact on wildlands.

23. They will need help: training, marketing support, and fertilizers, pesticides, and tools they can afford. This will require the full support of governments, including ensuring that conservation policies are designed with the benefit of agriculture foremost in mind. It may be expedient to stress the value to farmers rather than to wildlife of this programme, but in fact the destinies of the two are intertwined. Species conservation is tied to development, and the problems of both are more political than technical.

24. Population growth is a major threat to conservation efforts in many developing nations. Kenya has allocated 6 per cent of its territory as parks and reserves in order to protect its wildlife and to earn foreign exchange through tourism. But Kenya's present population of 20 million people is already pressing so hard on parks that protected land is steadily being lost to invading farmers. And the country's population is projected to grow fourfold in the next 40 years./10

25. Similar population pressures threaten parks in Ethiopia, Uganda. Zimbabwe, and other countries in which a growing but impoverished peasantry is forced to depend on a dwindling natural resource base. The prospects are bleak for parks that do not make important and recognizable contributions to national development objectives.

26. Brazil, Colombia, Indonesia. Cote d'Ivoire, Kenya, Madagascar, Peru, the Philippines. Thailand, and other nations with an unusual abundance of species already suffer a massive flow of farmers from traditional homelands into virgin territories. These areas often include tropical forests. perceived by the migrants encouraged to farm there as 'free' lands available for unimpeded settlement. The people who are already living on such lands at low population densities and with only traditional rights to the land are often swept aside in the rush to develop lands that might better be left in extensively used forest.

27. Many tropical countries with large forest resources have provoked wasteful 'timber booms' by assigning harvesting rights to concessionaires for royalty, rent, and tax payments that are only a small fraction of the net commercial value of the timber harvest. They have compounded the damage caused by these incentives by offering only short-term leases, requiring concessionaires to begin harvesting at once, and adopting royalty systems that induce loggers to harvest only the best trees while doing enormous damage to the remainder. In response, logging entrepreneurs in several countries have leased virtually the entire productive forest area

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within a few years and have overexploited the resource with little concern for future productivity (while unwittingly opening it for clearing by slash-and-burn cultivators)./11

28. In Central and South America, many governments have encouraged the large-scale conversion of tropical forests to livestock ranches. Many of these ranches have proved ecologically and economically unsound, as the underlying soils are soon depleted of nutrients; weed species replace planted grasses, and pasture productivity declines abruptly. Yet tens of millions of hectares of tropical forest have been lost to such ranches, largely because governments have underwritten the conversions with large land grants, tax credits and tax holidays, subsidized loans, and other inducements./12

29. The promotion of tropical timber imports into certain industrial countries, through low tariffs and favourable trade incentives, combined with weak domestic forest policies in tropical countries and with high costs and disincentives to harvesting in industrial countries, also drives deforestation. Some industrial countries typically import unprocessed logs either duty-free or at minimal tariff rates. This encourages developed country industries to use logs from tropical forests rather than their own, a pattern that is reinforced by domestic restrictions on the amounts that can be cut in domestic forests.

All of us in Africa are slowly waking up to the fact the African crisis is essentially an environmental problem that has precipitated such adverse symptoms as drought, famine, desertification, overpopulation, environmental refugees, political instability, widespread poverty, etc.

We are awaking to the fact that if Africa is dying it is because her environment has been plundered, overexploited, and neglected.

Many of us in Africa are also waking up to the realization that no good Samaritans will cross the seas to come to save the African environment. Only we Africans can and should be sufficiently sensitive to the well-being of our environment.

Mrs. Rahab W. Mwatha
The Greenbelt Movement
WCED Public Hearing
Nairobi, 23 Sept 1986

IV. Economic Values at Stake

30. Species conservation is not only justified in economic terms. Aesthetic, ethical, cultural, and scientific considerations provide ample grounds for conservation. For those who demand an accounting, the economic values inherent in the genetic materials of species are alone enough to justify species preservation.

31. Today, industrialized nations record far greater financial benefits from wild species than do developing countries, though unrecorded benefits to people living in the tropical countryside can be considerable. But the industrial countries have the scientific and industrial capacity to convert the wild material for industrial and medical use. And they also trade a higher proportion of their agricultural produce than do developing nations. Northern crop breeders are increasingly dependent on genetic materials from wild relatives of maize and wheat, two crops that play leading roles in the international grain trade. The U.S. Department of Agriculture estimates that contributions from plant genetic material lead to increases in productivity that average around l per cent annually, with a farm-gate value of well over $1 billion (1980 dollars)./13

32. The U.S. maize crop suffered a severe setback in 1970. when a leaf fungus blighted croplands, causing losses to farmers worth more than $2 billion. Then fungus-resistant genetic material was found in genetic stocks that had originated in Mexico./14 More recently, a primitive species of maize was discovered in a montane forest of south-central Mexico./15 This wild plant is the most primitive known relative of modern maize and was surviving in only three tiny patches covering a mere four hectares in an area threatened with destruction by farmers and loggers. The wild species is a perennial; all other forms of maize are annuals. Its cross-breeding with commercial varieties of maize opens up the prospect that farmers could be spared the annual expense of ploughing and sowing, since the plant would grow again yearly of its own accord. The genetic benefits of this wild plant, discovered when not more than a few thousand last stalks remained, could total several thousand million dollars a year./16

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33. Wild species likewise contribute to medicine. Half of all prescriptions dispensed have their origins in wild organisms./17 The commercial value of these medicines and drugs in the United States now amounts to some $14 billion a year./18 Worldwide, and including non-prescription materials plus pharmaceuticals, the estimated commercial value exceeds $40 billion a year./19 Industry also benefits from wildlife./20

34. Wildlife-derived materials contribute gums, oils, resins, dyes, tanning, vegetable fats and waxes, insecticides, and many other compounds. Many wild plants bear oil-rich seeds that can help in the manufacture of fibres, detergents, starch, and general edibles. For instance, the Fevillea genus of rain-forest vines in western Amazonia bear seeds with such a high oil content that a hectare of such vines in an original forest could produce more oil than a hectare of commercial oil palm plantation./21

35. A few plant species contain hydrocarbons rather than carbohydrates./22 Certain of these plants can flourish in areas that have been rendered useless through such activities as strip-mining. Hence land that has been degraded by extraction of hydrocarbons such as coal could be rehabilitated by growing hydrocarbons on the surface. Moreover, unlike an oil well, a 'petroleum plantation' need never run dry.

36. The emerging field of genetic engineering, by which science devises new variations of life forms, does not render wild genes useless. In fact, this new science must be based on existing genetic material and makes such material even more valuable and useful. Extinction, according to Professor Tom Eisner of Cornell University, 'no longer means the simple loss of one volume from the library of nature. It means the loss of a loose-leaf book whose individual pages, were the species to survive, would remain available in perpetuity for selective transfer and improvement of other species./23 And Professor Winston Brill of the University of Wisconsin has noted: 'We are entering an age in which genetic wealth, especially in tropical areas such as rain forests, until now a relatively inaccessible trust fund, is becoming a currency with high immediate value./24

37. Genetic engineering may mean that agriculture's Green Revolution will be superseded by a 'Gene Revolution'. This technology raises hopes of eventually harvesting crops from deserts, from seawater, and from other environments that did not previously support farming. Medical researchers foresee their own Gene Revolution bringing more innovative advances during the last two decades of this century than occurred during the previous 200 years.

38. Many of the nations with the least capacity for managing living resources are those richest in species; the tropics, which contain at least two-thirds of all species and a still greater proportion of threatened species, roughly coincide with the area generally referred to as the Third World. Many developing nations recognize the need to safeguard threatened species but lack the scientific skills, institutional capacities, and funds necessary for conservation. Industrial nations seeking to reap some of the economic benefits of genetic resources should support the efforts of Third World nations to conserve species; they should also seek ways to help tropical nations and particularly the rural people most directly involved with these species realize some of the economic benefits of these resources.

It will not be possible to restore the population of 'oomurasaki' - our purple emperor butterfly - to the previous level. The forest for oomurasaki requires weeding, planting of trees, and care and maintenance. The forest will be handed down to the succeeding generations. Isn't it wonderful to think that you are linked to the succeeding generations by handing down the forest where many oomurasaki fly and people enjoy themselves?

It would be nice if we could develop into the hearts of the children the love and affection for nature. We hope to make the forest we are making our gift to the children who will live in the 21st century.

Mika Sakakibara
Student, Tokyo University of Agriculture and Technology
WCED Public Hearing
Tokyo. 27 Feb 1987

V. A New Approach; Anticipate and Prevent

39. The historical approach of establishing national parks that are somehow isolated from the greater society has been overtaken by a new approach to conservation of species and ecosystems that can be characterized as 'anticipate and prevent.' This involves adding a new dimension to the now-traditional and yet viable and necessary step of protected areas. Development patterns must be altered to make them more compatible with

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the preservation of the extremely valuable biological diversity of the planet. Altering economic and land use patterns seems to be the best long-term approach to ensuring the survival of wild species and their ecosystems.

40. This more strategic approach deals with the problems of species depletion at their sources in development policies. anticipates the obvious results of the more destructive policies, and prevents damage now. A useful tool in promoting this approach is the preparation of National Conservation Strategies (NCS), which bring the processes of conservation and development together. Preparing an NCS involves government agencies, non-governmental organizations, private interests, and the community at large in analysis of natural resource issues and assessment of priority actions. In this way. it is hoped that sectoral interests will better perceive their interrelationships with other sectors and new potentials for conservation and development will be revealed.

41. The link between conservation and development and the need to attack the problem at the source can be seen clearly in the case of tropical forests. Sometimes it is government policy, not economic necessity, that drives the overexploitation and destruction of these resources. The direct economic and fiscal costs of this overexploitation - in addition to those of species extinction - are huge. The result has been wasteful exploitation of the tropical forests, the sacrifice of most of their timber and non-timber values. enormous losses of potential revenue to the government, and the destruction of rich biological resources.

42. Third World governments can stem the destruction of tropical forests and other reservoirs of biological diversity while achieving economic goals. They can conserve valuable species and habitat while reducing their economic and fiscal burdens. Reforming forest revenue systems and concession terms could raise billions of dollars of additional revenues, promote more efficient, long-term forest resource use. and curtail deforestation. Governments could save themselves enormous expense and revenue loss, promote more sustainable land uses, and slow down the destruction of tropical forests by eliminating incentives for livestock ranching.

43. The link between conservation and development also requires some changes in trade patterns. This has been recognized in the establishment in 1986 of the International Tropical Timber Organization, based in Yokohama, Japan, which seeks to rationalize trade flows. It has been set up to implement the first commodity agreement that incorporates a specific conservation component.

44. Numerous other opportunities can be found to encourage both species conservation and economic productivity. Many governments maintain unrealistically low taxes on rural land, while allowing settlers to establish title to 'virgin' land by converting it to farmland. Thus wealthy landowners can keep huge, underused estates at little or no cost, while land-hungry peasants are encouraged to clear forests to establish marginal holdings. Reforms of tax and tenure systems could increase productivity on existing holdings and reduce the pressures to expand cultivation into forests and upland watersheds.

45. Well-designed ecosystem conservation contributes to the predominant' goals of sustainable development in a number of ways. Safeguards for critical tracts of wildlands can serve also to safeguard agricultural land, for example. This is particularly true for upland forests of the tropics, which protect valley fields from floods and erosion, and waterways and irrigation systems from siltation.

46. A case in point is the Duraoga-Bone Reserve in Indonesia's northern Sulawesi, covering some 3,000 square kilometres of upland forest. It protects large populations of most of Sulawesi's endemic mammals, and many of the island's 80 endemic bird species. It also protects the Dumoga Valley Irrigation Scheme, funded by a World Bank loan, set up in the flatlands below to achieve a tripling of rice production on more then 13,000 hectares of prime agricultural land./25 Similar examples include the Canaima National Park in Venezuela, which protects domestic and industrial water supplies for a major hydropower facility that, in turn, provides electricity to the nation's key industrial centre and its capital city.

47. One conclusion from this connection is that governments could think of 'parks for development', insofar as parks serve the dual purpose of protection for species habitats and development processes at the same time. National efforts to anticipate and prevent the adverse consequences of development policies in any of these areas would surely yield much more for species conservation than all the measures of the past 10 years in support of park building, ranger patrols, anti-poaching units, and the other conventional forms of wildlife preservation. The 3rd World Congress on National Parks, held in Bali, Indonesia, in October 1982, brought this

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message from protected area managers to the policy makers of the world, demonstrating the many contributions that protected areas managed in the modern way are making to sustaining human society.

VI. International Action for National Species

48. Species and their genetic resources - whatever their origins - plainly supply benefits to all human beings. Wild genetic resources from Mexico and Central America serve the needs of maize growers and consumers globally. The principal cocoa-growing nations are in West Africa, while the genetic resources on which modern cocoa plantations depend for their continued productivity are found in the forests of western Amazonia.

49. Coffee growers and drinkers depend for the health of the crop on constant supplies of new genetic material from coffee's wild relatives, principally located in Ethiopia. Brazil, which supplies wild rubber germplasm to Southeast Asia's rubber plantation, itself depends on germplasm supplies from diverse parts of the world to sustain its sugar-cane, soybean, and other leading crops. Without access to foreign sources of fresh germplasm year by year, the nations of Europe and North America would quickly find their agricultural output declining.

50. The Earth's endowment of species and natural ecosystems will soon be seen as assets to be conserved and managed for the benefit of all humanity. This will necessarily add the challenge of species conservation to the international political agenda.

51. At the heart of the issue lies the fact that there is often a conflict between the short-term economic interest of the individual nations and the long-term interest of sustainable development and potential economic gains of the world community at large. A major thrust in actions to conserve genetic diversity must therefore be directed at making it more economically attractive both in the short term and in the longer perspective to protect wild species and their ecosystems. Developing countries must be ensured an equitable share of the economic profit from the use of genes for commercial purposes.

1. Some Current Initiatives

52. A number of international measures are already being tried. But they are limited in scope, only partially successful, and reactive in nature. UNESCO operates a clearing-house for information on natural areas and genetic resources. Its World Heritage Fund supports the management of a handful of exceptional ecosystems around the world, but all these activities receive small budgets. UNESCO has sought to establish a global system of Biosphere Reserves representing the Earth's 200 'biotic provinces' and harbouring sample communities of species. But only one-third of the needed reserves have been established, even though instituting and operating the rest would cost only about $150 million a year./26

53. UN agencies such as FAO and UNEP run programmes concerned with threatened species, genetic resources, and outstanding ecosystems. But their combined activities are tiny in the face of the large needs. Among national agencies, the U.S. Agency for International Development leads the field in recognizing the value of species conservation. Legislation passed by the U.S. Congress in 1986 will make available $2.5 million a year for this purpose./27 Again, this should be considered an important gesture compared with what has been done to date by bilateral agencies, but trifling compared with the needs and opportunities.

54. IUCN, working in close collaboration with UNEP, WWF, the World Bank, and various international technical assistance agencies, has established a 'Conservation Monitoring Centre', to provide data on species and ecosystems for any part of the world quickly and easily. This service, which is available to all, can help ensure that development projects are designed with full information available about the species and ecosystems that might be affected. Technical assistance is also available for nations, sectors, and organizations interested in establishing local data bases for their own applications.

55. Species problems tend to be perceived largely in scientific and conservationist terms rather than as a leading economic and resource concern. Thus the issue lacks political clout. One important initiative that attempts to put conservation more squarely on the agenda of international development concerns has been the Tropical Forestry Action Plan. This collaborative effort coordinated by FAO involves the World Bank, IUCN, the World Resources Institute, and UNDP, along with numerous other collaborating institutions. The broad-based effort proposes the formulation of national forestry reviews, national forestry plans, identification of new

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projects, enhanced cooperation between development aid agencies at work in the forestry sector. and increased flows of technical and financial resources into forestry and related fields such as' smallholder agriculture.

56. Establishing norms and procedures with respect to resource issues is at least as important as increased funding. Precedents for such norms include the Convention on Wetlands of International Importance, the Convention on Conservation of Islands for Science (both of which safeguard prime habitats and their species), and the Convention on International Trade in Endangered Species. These three precedents all help, although the first two are essentially reactive attempts to devise 'species refuges'.

> As deforestation progresses, it reduces the quality of life of millions of people in developing countries; their survival is threatened by the loss of the vegetation upon which they depend for their sources of household energy and many other goods. If tropical forests continue to be cleared at the current rate, at least 556 million acres (225 million hectares) will be cleared by the year 2000; if destruction of the tropical rain forests continues unabated, an estimated 10 to 20 per cent of the earth's plant and animal life will be gone by the year 2000.
>
> Reversing deforestation depends on political leadership and appropriate policy changes by developing-country governments to support community-level initiatives. The key ingredient is active participation by the millions of small farmers and landless people who daily use forests and trees to meet their needs.
>
> J. Gustave Speth
> President, World Resources Institute
> WCED Public Hearing
> Sao Paulo. 28-29 Oct 1985

2. Setting Priorities

57. A first priority is to establish the problem of disappearing species and threatened ecosystems on political agendas as a major resource issue. The World Charter for Nature, adopted by the UN in October 1982, was an important step towards this objective.

58. Governments should investigate the prospect of agreeing to a 'Species Convention', similar in spirit and scope to the Law of the Sea Treaty and other international conventions reflecting principles of 'universal resources'. A Species Convention, such as a draft prepared by IUCN, should articulate the concept of species and genetic variability as a common heritage.

59. Collective responsibility for the common heritage would not mean collective international rights to particular resources within nations. This approach need not interfere with concepts of national sovereignty. But it would mean that individual nations would no longer be left to rely on their own isolated efforts to protect species within their borders.

60. Such a Convention would need to be supported by a financial arrangement that would have the active backing of the community of nations. Any such arrangement, and there are several possibilities, must not only seek to ensure the conservation of genetic resources for all people, but assure that the nations that possess many of these resources obtain an equitable share of the benefits and earnings derived from their development. This would greatly encourage the conservation of species. One such arrangement might be a Trust Fund to which all nations could contribute, with those benefiting most from the use of these resources contributing an appropriate share. Governments of tropical forest nations could receive payments to support the conservation of given areas of forest, with such payments rising or falling depending on the degree to which the forests are maintained and protected./28

61. The sums required for effective conservation are large. Traditional-type conservation needs in tropical forests alone require outlays of $170 million a year for at least five years./29 However, the network of protected areas that the world will need by the year 2050 must include much larger areas brought under some degree of protection and a sophisticated degree of flexibility in management techniques./30

62. More funds will also be required for conservation activities outside protected areas: wildlife management, ecodevelopment areas, education campaigns, and so on. Other approaches of a less expensive sort include the conservation of wild gene reservoirs of special significance through 'genetic conservation areas' in countries

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well endowed with biological wealth. Much of this work can be carried out by citizens' groups and other non-governmental means.

63. International development agencies - the World Bank and other major lending banks. UN agencies, and bilateral agencies -should give comprehensive and systematic attention to the problems and opportunities of species conservation. Although the international trade in wildlife and wildlife products is considerable, to date the economic values inherent in genetic variability and ecological processes have been generally disregarded. Possible measures include environmental impact analyses of development projects with particular attention to species' habitats and life-support systems, identification of crucial localities featuring exceptional concentrations of species with exceptional levels of endemism that face exceptional degrees of threat, and special opportunities for linking species conservation with development aid.

VII. Scope for National Action

64. As indicated earlier, governments need to follow a new approach in this field - one of anticipating the impact of their policies in numerous sectors and acting to prevent undesirable consequences. They should review programmes in areas such as agriculture, forestry, and settlements that serve to degrade and destroy species' habitats. Governments should determine how many more protected areas are needed, especially in the spirit of how such areas can contribute to national development objectives, and make further provision for protection of gene reservoirs (for instance, primitive cultivated varieties) that may not normally be preserved through conventional protected areas.

65. In addition, governments need to reinforce and expand existing strategies. Urgent needs include better wildlife and protected-area management, more protected areas of a non-conventional type (such as the ecological stations that are proving reasonably successful in Brazil), more game cropping and ranching projects (such as the crocodile schemes in India, Papua New Guinea, Thailand, and Zimbabwe), more promotion of wildlife-based tourism, and stronger anti-poaching measures (even though relatively few species are threatened by poaching, compared with the vast numbers threatened by habitat loss). National Conservation Strategies, such as those already prepared in over 25 countries, can be important tools for coordinating conservation and development programmes.

66. Other measures governments could take to confront the crisis of disappearing species, recognizing that it constitutes a major resource and development challenge, include consideration of species conservation needs and opportunities in land use planning and the explicit incorporation of their genetic resource stocks into national accounting systems. This could entail establishing a natural-resource accounting system that directs particular attention to species as high-value yet little-appreciated resources. Finally, they should support and expand programmes of public education to ensure that the species question receives the attention it deserves throughout the entire population.

67. Every nation has only limited resources at its disposal for dealing with conservation priorities. The dilemma is how to use these resources most effectively. Cooperation with neighbouring nations sharing species and ecosystems can help streamline programmes as well as share expenses for regional initiatives. Explicit efforts to save particular species will be possible for only relatively few of the more spectacular or important ones. Agonizing as it will be to make such choices, planners need to make conservation strategies as systematically selective as possible. No one cares for the prospect of consigning threatened species to oblivion. But insofar as choices are already being made, unwittingly, they should be made with selective discretion that takes into account the impact of the extinction of a species upon the biosphere or on the integrity of a given ecosystem.

68. But even though public effort may be concentrated on a few species, all species are important and deserve some degree of attention; this might take the form of tax credits to farmers willing to maintain primitive cultivars. an end to incentives to clear virgin forest, the promotion of research attention from local universities, and the preparation of basic inventories of native flora and fauna by national institutions.

The world is unfortunately not what we would like it to be. The problems are many and great. Actually, they can only be solved with cooperation and quick-wittedness.
I represent an organization called 'Nature and Youth'. I know that I have full support among our members when I say that we are worried about the future if drastic changes do not take place, concerning the world's way of treating our essential condition, nature.

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We who work with youth, and are youth ourselves in Norway today, know very well how the destroying of nature leads to an apathetic fear among youth concerning their future and how it will turn out.

It is of great importance that common people get the chance to take part in deciding how nature should be treated.

Frederic Hauge
Nature and Youth
WCED Public Hearing
Oslo, 24-25 June 1985

VIII. The Need for Action

69. There are numerous signs that the loss of species and their ecosystems is being taken seriously as a phenomenon that carries practical implications for people all around the world, now and for generations to come.

70. The recent rise in public concern can be seen in such developments as the growth in Kenya's Wildlife Clubs. now numbering more than 1,500 school clubs with around 100.000 members./31 A parallel development in conservation education has occurred in Zambia. In Indonesia, some 400 conservation groups have joined together under the banner of the Indonesian Environmental Forum and exert strong political influence./32 In the United States, membership of the Audubon Society reached 385,000 in 1985./33 In the Soviet Union, nature clubs have over 35 million members./34 All of these indicate that the public puts a value on nature that is beyond the normal economic imperatives.

71. In response to this popular concern, governments have been moving to help species threatened within their borders, primarily through the establishment of additional protected areas. Today, the worldwide network of protected areas totals more than 4 million square kilometres, roughly equivalent to the size of most of the countries of Western Europe combined, or twice the size of Indonesia. In terms of continental coverage, protected areas in Europe (outside the USSR) amounted by 1985 to 3.9 per cent of territory; in the USSR, to 2.5 per cent; in North America, to 8.1 per cent; in South America, to 6.1 per cent; in Africa, to 6.5 per cent; and in Asia (outside the USSR) and Australia, to 4.3 per cent each./35

72. Since 1970, the networks have expanded in extent by more than 80 per cent, around two-thirds of which are in the Third World. But a great deal more remains to be done; a consensus of professional opinion suggests that the total expanse of protected areas needs to be at least tripled if it is to constitute a representative sample of Earth's ecosystems./36

73. There is still time to save species and their ecosystems. It is an indispensable prerequisite for sustainable development. Our failure to do so will not be forgiven by future generations.

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